Final Report

Exercise 2: Neural Quantum States and Sampled Renyi Entropy

This section introduces the neural-quantum-state workflow used later in the project. The model module contains the variational ansaetze, the sampler module generates configurations with a local Metropolis chain, and the observables module evaluates quantities such as the SWAP-based Renyi entropy. This separation is a deliberate design choice: once a model returns a wavefunction amplitude for a spin configuration, the same sampling and measurement layer can be reused across all three architectures.

2/a Neural Quantum State Representation

The common abstraction used throughout this exercise is a quantity of the form

$$\log \psi_\theta(\sigma), \tag{1}$$

which is the complex logarithm of the wavefunction amplitude for a spin configuration σ. This is the key interface boundary in the implementation. Once a model provides this object, the same sampling and observable code can be reused without model-specific logic in the Monte Carlo or SWAP-estimator layers.

All three ansatz classes first convert the stored spin variables from $\{0, 1\}$ to the more standard $\{-1, +1\}$ convention. The final network output is then interpreted as two real channels: one for the log-amplitude and one for a raw phase variable $p_\theta(\sigma)$. In the implemented models the phase is bounded as $\phi_\theta(\sigma) = \pi \tanh p_\theta(\sigma)$, so the output takes the form

$$\log \psi_\theta(\sigma) = a_\theta(\sigma) + i\pi \tanh p_\theta(\sigma). \tag{2}$$

This keeps the interface uniform across all architectures and makes it straightforward to evaluate amplitude ratios later in the sampler and in the SWAP estimator.

For the present TFIM benchmark a real non-negative representation would in principle be sufficient in the computational basis, because the model is stoquastic there and its ground state can be chosen positive by a Perron–Frobenius argument.[1] The project still keeps a complex-valued interface because the same code is meant to carry over to frustrated models, where a positive real representation is not generally available. In particular, for the square-lattice J_1-J_2 antiferromagnet the Marshall–Peierls sign rule is known to break down once the frustration is strong enough, so the wavefunction cannot in general be treated as an everywhere-positive amplitude in the computational basis.[2]

The RBM ansatz is the closest to the standard NQS construction. Its hidden width is set by an integer density parameter α, so the number of hidden units scales as αN. After the visible

[1] See Bravyi et al., "Monte Carlo simulation of stoquastic Hamiltonians," *Quantum Information and Computation* 15 (2015), which explicitly lists the transverse-field Ising model as a stoquastic example.

[2] See Bishop, Farnell, and Parkinson, "Phase Transitions in the Spin-Half J_1–J_2 Model," *Phys. Rev. B* 58 (1998), which reports a breakdown of the Marshall–Peierls sign rule on the square lattice at $J_2/J_1 \approx 0.26$.

spins are mapped to $\{-1, +1\}$, the hidden variables are summed out analytically, which gives the usual closed-form log-amplitude,

$$\log \psi_\theta(\sigma) = \sum_i a_i \sigma_i + \sum_j \log\cosh(W_j \cdot \sigma + b_j) + i\pi \tanh\left[\sum_i \tilde{a}_i \sigma_i + \sum_j \left(\tilde{W}_j \cdot \sigma + \tilde{b}_j\right)\right]. \quad (3)$$

In the implemented RBM, the amplitude channel therefore keeps the standard analytically summed hidden contribution, while a separate linear phase channel supplies the complex phase before the final bounding step. Of the three models, this is the one that stays closest to the original RBM-NQS construction.

The FFNN ansatz replaces the analytic RBM structure by a dense multilayer perceptron. A configurable sequence of hidden widths defines the network depth and capacity, each hidden layer is followed by a fixed activation, and the final dense layer produces the two output channels for log-amplitude and phase. Unlike the RBM, there is no special hidden-spin summation here; the dependence on the spin configuration is learned entirely through the stacked nonlinear dense layers.

The CNN ansatz keeps the same output convention but changes the internal representation of the state. Instead of treating the spin configuration as a flat vector throughout, it is first reshaped into a lattice-shaped tensor, then passed through a stack of convolution layers, and finally flattened again before the two-channel output head. This is the only ansatz that explicitly takes the lattice geometry into account, so it can encode local geometric structure and translation-related patterns more naturally than a dense network.

2/b Architecture-Agnostic Monte Carlo Sampling

Sampling is performed with a local Metropolis scheme acting directly on spin configurations. Starting from a batch of chains, one site is chosen uniformly at random in each chain and the corresponding spin is flipped, producing a proposed state σ'. The acceptance probability is computed from the wavefunction ratio,

$$A(\sigma \to \sigma') = \min\left(1, \frac{|\psi_\theta(\sigma')|^2}{|\psi_\theta(\sigma)|^2}\right) = \min\left(1, \exp\left[2\,\mathrm{Re}\big(\log\psi_\theta(\sigma') - \log\psi_\theta(\sigma)\big)\right]\right). \quad (4)$$

This is exactly the expression used in the implementation: the log-amplitudes of the current and proposed states are evaluated in batch, and only their real parts enter the Metropolis ratio because the sampling measure is $|\psi_\theta(\sigma)|^2$.

The same sampler is used for RBM, FFNN, and CNN. In particular, no stochastic reconfiguration is introduced for the RBM in this exercise, for the sake of a unified VMC scheme for all three ansaetze.

The proposal move is a single-spin local update. This is the simplest option and the cleanest common baseline because it does not depend on any architecture-specific structure or model-specific proposal design. It also matches the project interface naturally: once $\log\psi_\theta(\sigma)$ is available, a local flip proposal can be scored without any extra model logic. In that sense it is the most transparent architecture-agnostic choice.

This simplicity comes with a cost. Alternatives such as multi-spin proposals, exchange moves, cluster-style updates, or more structured Hamiltonian-aware proposal rules could in principle improve mixing, especially when the state develops stronger correlations. A single-spin local move changes the configuration only minimally at each step, so one expects slower decorrelation when the correlation length becomes large. This is the usual critical-slowing-down problem of local Ising updates: near a continuous transition, increasingly large correlated regions have to be rearranged through many local moves.[3] Deep in the ordered phase the system is still strongly

[3] This is the motivation behind cluster algorithms such as Swendsen–Wang; see Swendsen and Wang, "Nonuniversal critical dynamics in Monte Carlo simulations," *Phys. Rev. Lett.* 58 (1987).

correlated, but it is no longer critical in the same sense, so the correlation length is finite and the slowdown is less severe than in the near-critical regime. In the present implementation, acceptance is therefore supplemented by a magnetization-autocorrelation diagnostic. The code does not yet estimate an integrated autocorrelation time, so acceptance alone should not be read as a full measure of sampler quality.

Figure 1 should be read in that limited sense. It shows that the sampler is numerically stable for the runs shown, but it is not by itself a full mixing study.

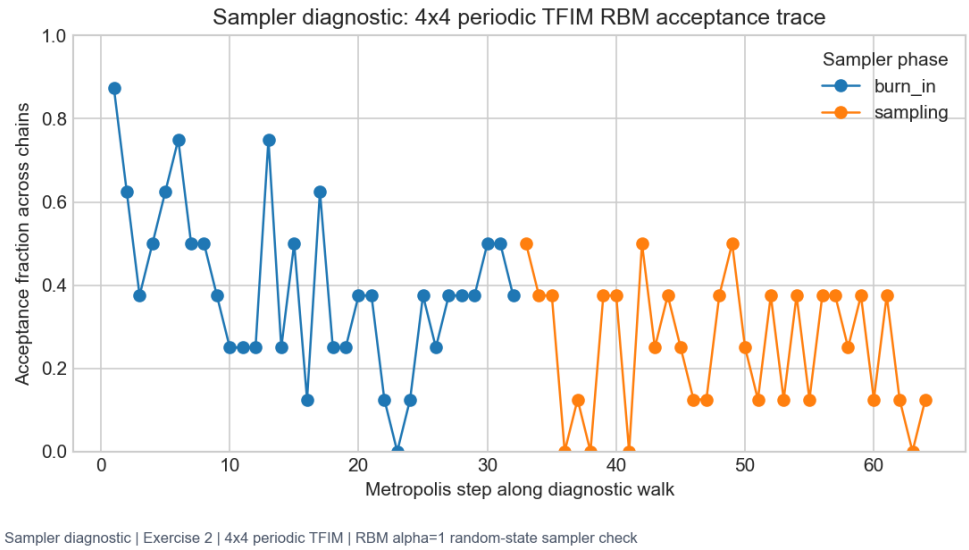

Figure 1: Sampler acceptance for representative Exercise 2 runs.

2/c Renyi-2 From the SWAP Estimator

The practical entanglement observable carried over from Exercise 1 is the Renyi-2 entropy,

$$S_2(A) = -\log \operatorname{tr}(\rho_A^2). \tag{5}$$

For sampled neural quantum states, the useful identity is the replica-SWAP expression[4]

$$\operatorname{tr}(\rho_A^2) = \langle \mathrm{SWAP}_A \rangle, \tag{6}$$

which can be estimated from pairs of independently sampled configurations. Writing two configurations as $\sigma = (\sigma_A, \sigma_B)$ and $\eta = (\eta_A, \eta_B)$, the local SWAP estimator is

$$\mathrm{SWAP}_A^{\mathrm{loc}}(\sigma, \eta) = \frac{\psi_\theta(\eta_A, \sigma_B)\, \psi_\theta(\sigma_A, \eta_B)}{\psi_\theta(\sigma_A, \sigma_B)\, \psi_\theta(\eta_A, \eta_B)}. \tag{7}$$

Taking the average of this quantity over sampled replica pairs gives an estimator for $\langle \mathrm{SWAP}_A \rangle$, and therefore for $S_2(A)$.

The implementation follows this formula directly. Samples are flattened into a batch, paired row by row, and the subsystem-A spins are exchanged between the two replicas to form the swapped configurations. The code then evaluates the log-amplitudes of the swapped states and constructs the estimator through exponentiated log-amplitude differences,

$$\mathrm{SWAP}_A^{\mathrm{loc}}(\sigma, \eta) = \exp\Big[\log \psi_\theta(\eta_A, \sigma_B) + \log \psi_\theta(\sigma_A, \eta_B) - \log \psi_\theta(\sigma_A, \sigma_B) - \log \psi_\theta(\eta_A, \eta_B)\Big]. \tag{8}$$

[4]For the original quantum Monte Carlo formulation of measuring Renyi entropy through a SWAP operator, see Hastings, Gonzalez, Kallin, and Melko, "Measuring Renyi Entanglement Entropy in Quantum Monte Carlo Simulations," *Phys. Rev. Lett.* 104 (2010).

This is exactly the form that makes the shared $\log \psi_\theta$ interface useful: the observable code only needs log-amplitude evaluations on original and swapped configurations, so no model-specific entanglement logic is required.

The small-system validation is summarized in Table 1 and Figure 2. Here the exact $S_2(A)$ values come from explicit state reconstruction on the benchmark system, while the sampled $S_2(A)$ values are Monte Carlo estimates obtained from the SWAP estimator. The comparison therefore tests whether the sampled route reproduces the exact benchmark within the available noise level.

$\lvert A\rvert$	exact $S_2(A)$	sampled $S_2(A)$	sampled std	estimator std
1	0.2246	0.2115	0.1188	0.0336
2	0.3212	0.3264	0.1901	0.0430

Table 1: SWAP-validation data against exact small-system benchmark values.

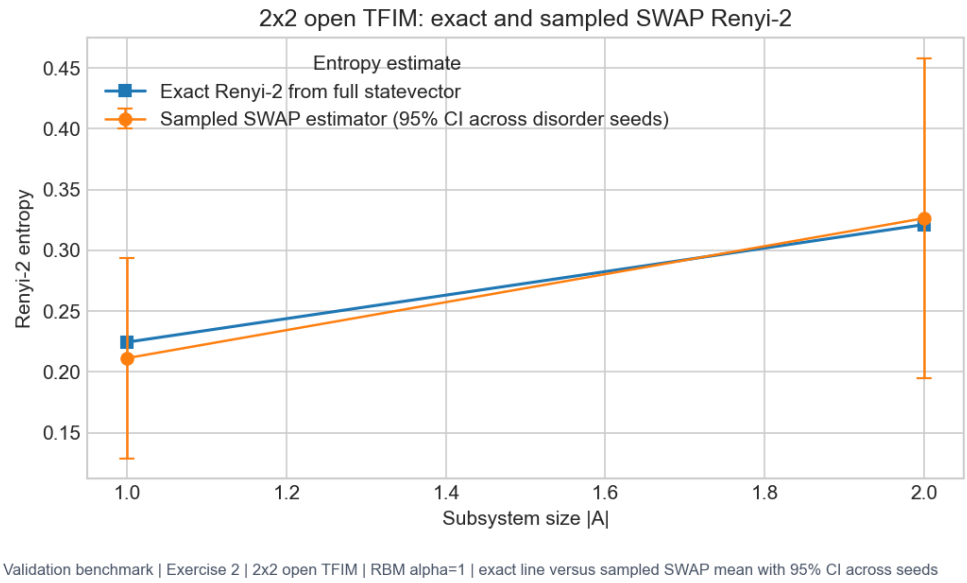

Figure 2: Sampled SWAP estimates compared with exact Renyi-2 benchmark values.

The agreement is good at the level expected from the current Monte Carlo noise. For both retained subsystem sizes the sampled mean lies close to the exact value, with the remaining discrepancy at $|A| = 1$ still small compared with the observed sampling spread. This is enough to show that the implementation is targeting the correct quantity, while still making clear that the estimator is not quantitatively sharp without a larger sample budget.

That same point explains the noise discussion. As the subsystem grows, the swapped configurations become less typical under the original sampling distribution, so the ratio in the local estimator fluctuates more strongly. In practice this means that larger subsystems tend to produce noisier SWAP estimates. Higher-order Renyi entropies could be defined in the same replica spirit, but they would require more replicas and typically inherit even worse estimator quality. In the present setting, Renyi-2 is therefore the practical compromise between physical usefulness and estimator stability.

2/d Random Initialization Choices

The first random-state study asks how strongly the sampled half-partition Renyi-2 depends on initialization before any training is performed. The two relevant choices are the overall parameter

scale and whether the phase channel is allowed to fluctuate freely from the start. The summary is given in Table 2 and Figure 3.

initialization	valid fraction	sampled $S_2(A)$	sampled std	estimator std
default complex	0.00	—	0.0000	0.0000
real scale 0.25	1.00	0.0409	0.0407	0.0415
real scale 1	1.00	1.4315	0.5413	0.8928
real scale 2	1.00	4.1525	1.5056	2.6525

Table 2: Summary of the random-initialization study.

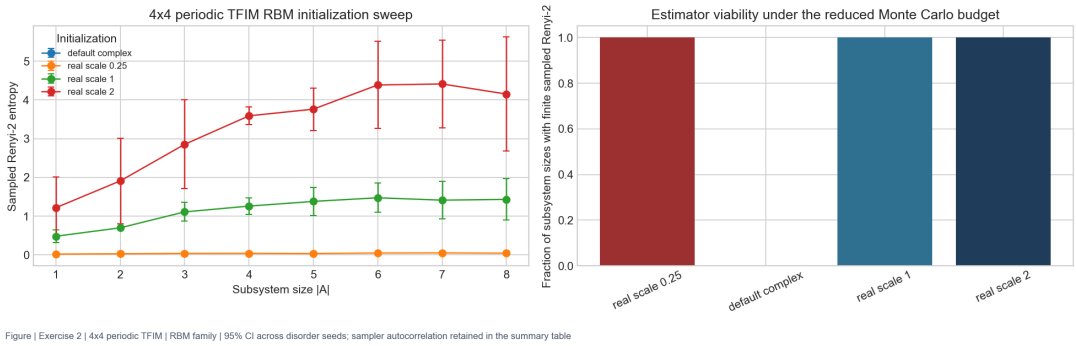

Figure 3: Initialization comparison for the periodic 4×4 TFIM random-state study.

The clearest conclusion is that unrestricted random complex phases are not usable under the reduced initialization-study budget. The default complex initialization produces no valid entropy points in the summary table, so allowing the phase channel to fluctuate freely from the outset pushes the SWAP estimator outside its numerically stable regime. This is consistent with the structure of the estimator: once the phase varies too strongly between original and swapped configurations, the replica ratio becomes much harder to average reliably.

Within the real-amplitude family, the overall scale matters strongly. A small real scale gives a state very close to a low-entanglement random baseline, while larger scales produce substantially larger sampled Renyi-2 values even before any optimization. In these data the half-partition estimate rises from about 0.041 at scale 0.25 to about 1.43 at scale 1 and about 4.15 at scale 2, although the uncertainty also grows sharply with scale. The interpretation is not that larger scales are automatically better, but that the random-state entanglement seen by the sampler is already highly sensitive to how broadly the amplitude channel is initialized.

2/e Architecture Comparison

The second random-state study compares architecture families at a fixed real-amplitude initialization rule. The summary is listed in Table 3 and shown in Figure 4. Because the comparison is performed before optimization, it is best read as a comparison of representational and sampling behavior under a common initialization rule rather than as a definitive ranking of architectures.

model	sampled $S_2(A)$	sampled std	estimator std
CNN channels=4	0.0030	0.0021	0.0077
CNN channels=6	0.0037	0.0022	0.0093
FFNN width=8	0.0229	0.0129	0.0231
FFNN width=12	0.0320	0.0122	0.0320
RBM alpha=1	1.4854	0.2254	0.6003
RBM alpha=2	1.8217	0.3548	1.0675

Table 3: Summary of the architecture-comparison study.

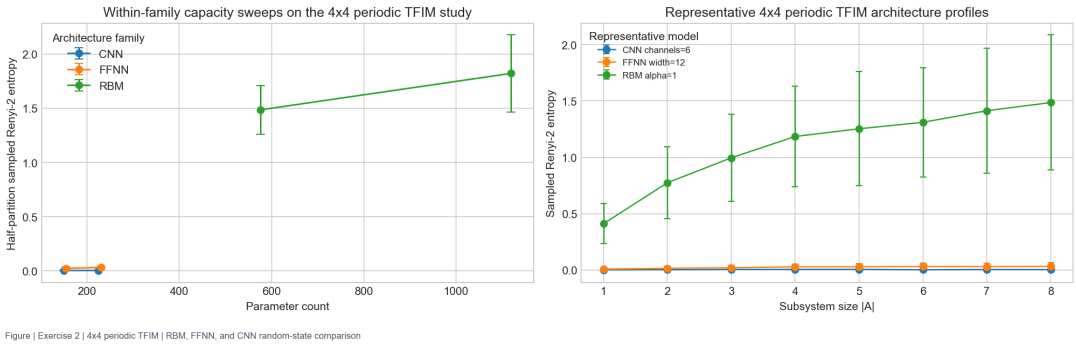

Figure 4: Architecture comparison for the periodic 4×4 TFIM random-state study.

The largest separation in the table is between the RBM and the other two architectures: the RBM produces order-one half-partition Renyi-2 values, whereas the FFNN and CNN remain close to zero. That observation is useful as a practical benchmark, but it should not be overinterpreted as a physics conclusion. Under the present initialization and sampling budget, the comparison mainly shows that the RBM explores substantially more entangled random states than the FFNN and CNN.

The within-family trends are modest for the FFNN and CNN families but clearer for the RBM. Increasing the FFNN width only changes the half-partition entropy slightly, and the two CNN choices remain close to the low-entanglement baseline. By contrast, the larger RBM reaches the highest sampled half-partition Renyi-2 in the comparison, which is why Exercise 3 carries forward the higher-capacity RBM variant.

Conclusion

Exercise 2 sets up the neural-quantum-state side of the project around a common $\log \psi_\theta(\sigma)$ interface. That choice makes it possible to compare an RBM, a dense feed-forward network, and a convolutional network without changing the surrounding Monte Carlo and observable code. The local Metropolis sampler is kept deliberately simple and uniform across the three ansatz families, even though that simplicity comes with the expected weakness that single-spin updates decorrelate slowly as correlations grow.

The practical entanglement observable is the sampled Renyi-2 entropy obtained from the SWAP estimator. The small-system benchmark compares that sampled estimator directly against exact Renyi-2 values and shows that the method targets the correct observable, while also making clear that finite-sample noise is already a real issue even in a favorable setting. That is precisely why Renyi-2 is the natural quantity to carry forward: it remains accessible from samples, whereas more demanding entanglement probes would be even harder to estimate reliably.

The random-state studies make two additional points. First, initialization matters strongly, especially through parameter scale and through whether the phase channel is allowed to fluctuate freely. Second, complex amplitudes are not merely an implementation convenience: they are needed if the same framework is to extend beyond stoquastic benchmarks to frustrated models such as square-lattice J_1-J_2 systems, where a globally positive real representation is not generally available. Taken together, these results establish the measurement route used later in the project and clarify the practical constraints under which sampled entanglement estimates remain reliable.